other species of particles (such as anti-electrons and particles known as pions). But after years of painstaking search for such *proton decay* in elaborate underground experiments—the experiment Georgi had excitedly described to me in his office years ago—none were found; this ruled out Georgi and Glashow's proposal. Since then, however, physicists have developed variations on that original model that are not ruled out by such experiments; however, so far none of these alternative theories have been confirmed.

The consensus among physicists is that grand unification is one of the great, as yet unrealized, ideas in particle physics. Since unification and cosmological phase transitions have proven so potent for electromagnetism and the weak nuclear force, many feel that it is only a matter of time before other forces are also gathered within a unified framework. As we shall see in Chapter 12, great strides in this direction have recently been made using a different approach—*superstring theory*—that has, for the first time, brought *all* forces, including gravity, into a unified theory, albeit one which is still, as of this writing, under vigorous development. But what is already clear, even in just considering the electroweak theory, is that the universe we currently see exhibits but a remnant of the early universe's resplendent symmetry.

The Return of the Aether

The concept of symmetry's breaking, and its realization through the electroweak Higgs field, clearly plays a central role in particle physics and cosmology. But the discussion may have left you wondering about the following: If a Higgs ocean is an invisible something that fills what we ordinarily think of as empty space, isn't it just another incarnation of the long discredited notion of the aether? The answer: yes and no. The explanation: yes, indeed, in some ways a Higgs ocean does smack of the aether. Like the aether, a condensed Higgs field permeates space, surrounds us all, seeps right through everything material, and, as a nonremovable feature of empty space (unless we reheat the universe above 10^{15} degrees, which we can't actually do), it redefines our conception of nothingness. But unlike the original aether, which was introduced as an invisible medium to carry light waves in much the same